F18CF – Linear Algebra

Tutorial Sheet 9: Diagonalisability, Dynamical Systems

Getting started.

1. Find the eigenvalues and a basis of the eigenspaces of the following matrices:

$$\begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}, \quad \begin{pmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}, \quad \begin{pmatrix} 2 & -2 & 3 \\ 10 & -4 & 5 \\ 5 & -4 & 6 \end{pmatrix}.$$

   In each case determine whether the matrix is diagonalisable.

2. We consider the following matrix:

$$A = \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix}.$$

   (a) Show that A is diagonalisable, and compute an invertible 2×2 matrix P and diagonal 2×2 matrix such that $A = PDP^{-1}$.

   (b) Compute A^{10}.

Problems.

3. **A prey-predator system.** We wish to study the long-term evolution of a population of frogs and gnats living by a large pond. At the beginning of the study, there are $f_0 = 100$ frogs and $g_0 = 5000$ gnats. We denote by f_n the number of frogs after n years, and by g_n the number of gnats after n years. We assume that the following simplified model holds:

$$\begin{cases} f_{n+1} &= 0.5f_n + 0.3g_n \\ g_{n+1} &= -0.2f_n + 1.2g_n \end{cases}$$

   (a) What is the expected population after one year? After two years?

   (b) Write down the matrix A associated to this dynamical system and compute its eigenvalues λ_1, λ_2. Show that A is diagonalisable.

   (c) Using the fact that the initial condition vector $(f_0, g_0)^T \in \mathbb{R}^2$ can be written in a basis of $\mathbb{R}^2$ made of eigenvectors of A, show that there exist constants $a_1, a_2, b_1, b_2 \in \mathbb{R}$ such that for every $n \geq 0$, we have:

$$\begin{cases} f_n &= a_1\lambda_1^n + a_2\lambda_2^n \\ g_n &= b_1\lambda_1^n + b_2\lambda_2^n \end{cases}$$

   (d) Compute the constants a_1, a_2, b_1, b_2, and deduce an exact formula for f_n and for g_n for every n.

(e) Show that the ratio g_n/f_n converges to a limit, and compute that stable ratio.

4. **A sequence satisfying a linear recurrence.** We consider the sequence u_n defined by $u_0 = 0, u_1 = 1, u_2 = 2$, and for $n \geq 0$:

$$u_{n+3} = u_{n+2} + \frac{1}{4}u_{n+1} - \frac{1}{4}u_n.$$

(a) For $n \geq 0$, we set $\vec{x}_n := (u_n, u_{n+1}, u_{n+2})^T \in \mathbb{R}^3$. Show that we have

$$\vec{x}_{n+1} = A\vec{x}_n$$

for an appropriate 3×3 matrix A.

(b) Compute the characteristic polynomial of A and the eigenvalues of A. (Hint: check that 1 is an eigenvalue.) Deduce that A is diagonalisable.

(c) We denote by $\lambda_1, \lambda_2, \lambda_3$ the three eigenvalues of A. Using the fact that the initial condition vector $\vec{x}_0 \in \mathbb{R}^3$ can be written in a basis of $\mathbb{R}^3$ made of eigenvectors of A, show that there exist constants $a_1, a_2, a_3 \in \mathbb{R}$ such that, for every $n \geq 0$,

$$u_n = a_1\lambda_1^n + a_2\lambda_2^n + a_3\lambda_3^n,$$

and compute a_1, a_2, a_3.

(d) What is the limit of u_n?

5. **Diagonalisability of real symmetric 2×2 matrices.** A square matrix is called **symmetric** if $A = A^T$. Let

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

be a real symmetric 2×2 matrix. In parts (a)-(c), we assume that $b \neq 0$.

(a) Compute the characteristic polynomial of A. Show that A has two distinct eigenvalues λ_1, λ_2, and deduce that A is diagonalisable.

(b) We consider the standard (Euclidean) inner product on $\mathbb{R}^2$. Show that $\langle A\vec{x}, \vec{y}\rangle = \langle \vec{x}, A\vec{y}\rangle$ for every $\vec{x}, \vec{y} \in \mathbb{R}^2$.

(c) Let $\vec{v}_1, \vec{v}_2$ be eigenvectors of A for the eigenvalues λ_1, λ_2 respectively. Show that $\vec{v}_1, \vec{v}_2$ form an orthogonal basis of $\mathbb{R}^2$.

(d) By also considering the case $b = 0$, deduce that for every real 2×2 symmetric matrix A there exists an orthogonal basis of $\mathbb{R}^2$ made of eigenvectors of A.

6. **An application to Multivariable Calculus.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a smooth function, and let $\vec{e}_1, \vec{e}_2$ be the standard orthogonal basis of $\mathbb{R}^2$. Suppose that $\vec{0} \in \mathbb{R}^2$ is a critical point of f, i.e. $\dfrac{\partial f}{\partial x_1}(\vec{0}) = 0$ and $\dfrac{\partial f}{\partial x_2}(\vec{0}) = 0$. You have seen in Multivariable Calculus that around the critical point $\vec{0}$, we have the following approximation for $\vec{x} \in \mathbb{R}^2$:

$$f(\vec{x}) = f(\vec{0}) + \frac{1}{2}\vec{x}^T H\vec{x} + \text{ (higher order terms)},$$

where the matrix H is the *Hessian* matrix of f, which is the 2×2 matrix whose (i,j)-entry is $\dfrac{\partial^2 f}{\partial x_i \partial x_j}(\vec{0})$. The Hessian matrix is symmetric by Schwarz's Theorem. Using Problem 5, we choose an orthonormal basis $\vec{v}_1, \vec{v}_2$ of eigenvectors of H, and let λ_1, λ_2 be the corresponding eigenvalues.

(a) Show that for a vector $\vec{x} \in \mathbb{R}^2$ written as $\vec{x} = a_1\vec{v}_1 + a_2\vec{v}_2$ (with $a_1, a_2 \in \mathbb{R}$), we have:

$$f(\vec{x}) = f(\vec{0}) + \frac{1}{2}\big(\lambda_1 a_1^2 + \lambda_2 a_2^2\big) + \text{ (higher order terms).}$$

(b) We suppose that H is invertible. Under which conditions on the signs of λ_1 and λ_2 is $\vec{0}$ a local maximal? A local minimal? A saddle point?

(c) Under which conditions on the signs of $\mathrm{Tr}(H)$ and $\det(H)$ is $\vec{0}$ a local maximal? A local minimal? A saddle point?